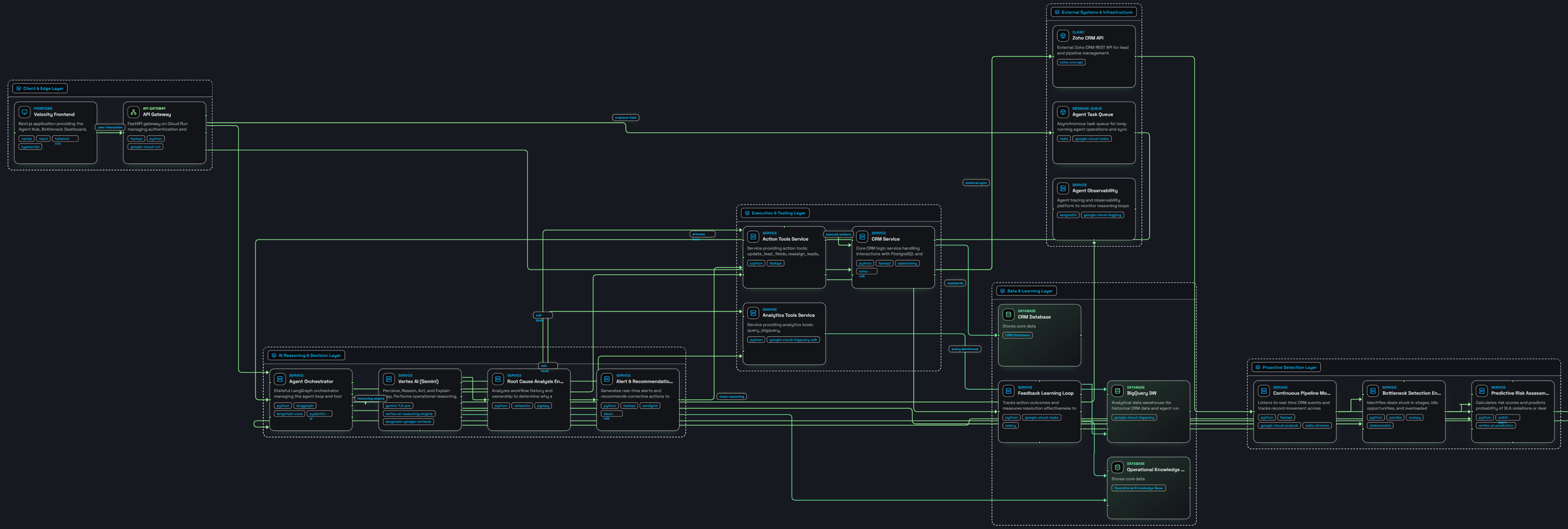
Client & Edge Layer
FRONTEND
Velocity Frontend
Next.js application providing the Agent Hub, Bottleneck Dashboard,
nextjs
react
tailwindcss
typescript
user interaction
API GATEWAY
API Gateway
FastAPI gateway on Cloud Run managing authentication and
fastapi
python
google-cloud-run
enqueue task
External Systems & Infrastructure
CLOUD
Zoho CRM API
External Zoho CRM REST API for lead and pipeline management.
zoho-crm-agt
MESSAGE-QUEUE
Agent Task Queue
Asynchronous task queue for long-running agent operations and sync
redis
google-cloud-tasks
SERVICE
Agent Observability
Agent tracing and observability platform to monitor reasoning loops
langsmith
google-cloud-logging
external sync
Execution & Tooling Layer
process task
SERVICE
Action Tools Service
Service providing action tools: update_lead_fields, reassign_leads,
python
fastapi
execute actions
SERVICE
CRM Service
Core CRM logic service handling interactions with PostgreSQL and
python
fastapi
sqlalchemy
zoho-crm
read/write
SERVICE
Analytics Tools Service
Service providing analytics tools: query_bigquery,
python
google-cloud-bigquery-sdk
call tools
query warehouse
Data & Learning Layer
DATABASE
CRM Database
Stores core data
CRM Database
SERVICE
Feedback Learning Loop
Tracks action outcomes and measures resolution effectiveness to
python
google-cloud-tasks
celery
DATABASE
BigQuery DW
Analytical data warehouse for historical CRM data and agent run
google-cloud-bigquery
DATABASE
Operational Knowledge ...
Stores core data
Operational Knowledge Base
AI Reasoning & Decision Layer
tool loops
SERVICE
Agent Orchestrator
Stateful LangGraph orchestrator managing the agent loop and tool
python
langgraph
langchain-core
pydantic-...
reasoning engine
SERVICE
Vertex AI (Gemini)
Perceive, Reason, Act, and Explain loop. Performs operational reasoning,
gemini-1.5-pro
vertex-ai-reasoning-engine
langchain-google-vertexai
SERVICE
Root Cause Analysis En...
Analyzes workflow history and ownership to determine why a
python
networkx
pgmpy
SERVICE
Alert & Recommendatio...
Generates real-time alerts and recommends corrective actions to
python
fastapi
sendgrid
slack-sdk
trace reasoning
Proactive Detection Layer
SERVICE
Continuous Pipeline Mo...
Listens to real-time CRM events and tracks record movement across
python
fastapi
google-cloud-pubsub
redis-streams
SERVICE
Bottleneck Detection En...
Identifies deals stuck in stages, idle opportunities, and overloaded
python
pandas
numpy
statsmodels
SERVICE
Predictive Risk Assessm...
Calculates risk scores and predicts probability of SLA violations or deal
python
scikit-...
vertex-ai-prediction